

Master's Thesis
**Multi-objective Optimisation of suction-cup and two
finger grippers using todo: <Algorithims>**

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Zitat

‘Far out in the uncharted backwaters of the unfashionable end of the western spiral arm of the Galaxy lies a small unregarded yellow sun. Orbiting this at a distance of roughly ninety-two million miles is an utterly insignificant little blue green planet whose ape-descended life forms are so amazingly primitive that they still think digital watches are a pretty neat idea.’

Douglas Adams – The Hitchhikers Guide to the Galaxy

Vorwort

Dank an die Firma und die Firmenmitarbeiter, max. 1/2 Seite

Kurz-Zusammenfassung

‘Aushängeschild’’ der Arbeit, max 1 Seite

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Nomenclature

(x_i, y_i)	Planar position of the i -th suction cup
$\boldsymbol{\theta}$	Algorithm hyperparameters
δ_{intra}	Spatial threshold for selecting additional local representatives
ℓ_i	Lower bound of the i -th decision variable
Λ	Set of reference directions for decomposition-based algorithms
$\mathbf{f}(\mathbf{p})$	Multi-objective function vector
$\mathbf{p}$	Decision variable vector representing the gripper configuration
$\mathbf{r}$	Reference point for hypervolume computation
$\mathcal{A}$	Selected multi-objective evolutionary algorithm
$\mathcal{F}^{(t)}$	Objective values of the population at generation t
$\mathcal{G}_{\text{cand}}$	Set of candidate grasps generated from sampled surface points
$\mathcal{G}_{\text{rep}}$	Reduced representative grasp set used for diversity evaluation
$\mathcal{G}_{\text{valid}}$	Set of valid (collision-free) grasps
$\mathcal{P}^*$	Set of Pareto-optimal solutions
$\mathcal{P}^{(0)}$	Initial population of candidate solutions
$\mathcal{S}$	Set of sampled surface points on the workpiece
$\mathcal{Z}$	Set of random seeds for meta-optimization
$\text{HV}^{(t)}$	Hypervolume indicator at generation t
$A(\mathbf{p})$	Area of the triangle formed by the three suction cups
$D(\mathbf{p})$	Diversity metric of the grasp distribution
$d_{\text{geo}}(\mathbf{R}_a, \mathbf{R}_b)$	Geodesic distance between two grasp orientations
$d_{\text{ori}}(i, j)$	Pairwise orientation distance between grasp poses i and j
$D_{\text{pos}}(\mathbf{p})$	Spatial component of the diversity metric
$d_{\text{pos}}(i, j)$	Pairwise translational distance between grasp poses i and j
$D_{\text{rot}}(\mathbf{p})$	Orientation component of the diversity metric
E_{max}	Maximum number of function evaluations

f_{tol}	Objective space convergence tolerance
G_{max}	Maximum number of generations
M	Number of initially sampled surface points
N	Population size
n	Number of decision variables
n_{obj}	Number of Objectives
q_{intra}	Maximum number of local orientation-diverse representatives
s	Characteristic object scale derived from the mesh bounding box
t	Generation index
u_i	Upper bound of the i -th decision variable
x_{tol}	Design space convergence tolerance

1 Introduction

1.1 Motivation

Robotic grasping is a fundamental capability in automated manufacturing, logistics, and service robotics[1]. The ability to reliably grasp objects of varying shapes, sizes, and surface properties remains a challenging problem, particularly in unstructured environments[2]. Industrial applications such as bin picking, assembly, and material handling require grippers that can operate robustly across a wide range of workpieces[3].

Among the commonly used end-effectors, suction-cup grippers and two-finger grippers represent two complementary approaches[4]. Suction grippers are widely used for handling objects with smooth surfaces due to their simplicity and speed, whereas two-finger grippers provide flexibility in handling objects with complex geometries. Designing optimal configurations for these grippers is non-trivial, as grasp performance depends on multiple interacting factors such as geometry, contact distribution, force limits, and collision constraints[5].

While the gripper configuration problems are well-motivated in themselves, the **primary objective of this thesis** is not simply to find the best gripper configuration, but to **systematically apply and compare optimization algorithms** for solving these configuration problems. A large number of optimization algorithms exist, spanning evolutionary strategies, decomposition-based methods, indicator-based selection, and Bayesian optimization, yet their relative suitability for gripper design problems of this type is not well understood. Different algorithms may exhibit substantially different convergence behaviour, solution quality, and robustness depending on the structure of the objective landscape and the dimensionality of the search space. Identifying which algorithm class performs best for each gripper problem is therefore a practically relevant and scientifically meaningful research question[6].

To provide a rigorous and reproducible basis for this comparison, two gripper configuration problems are formulated as well-defined multi-objective optimization tasks. In both cases, a diverse set of workpieces is incorporated directly into the problem formulation by treating each workpiece as a separate objective. This design choice ensures that the optimized gripper configurations are not tailored to a single geometry, but are instead robust across a range of workpiece shapes. It also increases the dimensionality of the objective space as the number of workpieces grows, allowing the benchmarking study to probe algorithm behaviour under varying levels of problem complexity[7].

For the **multi-suction-cup gripper**, the objectives consist of the geometric stability of the suction base together with the grasp diversity evaluated independently for each workpiece. These objectives are inherently conflicting: a larger suction base improves stability but increases the risk of collisions with the workpiece, reducing the number of feasible grasps.

For the **two-finger gripper**, each workpiece contributes one objective, representing how well

the optimized finger contour conforms to that particular workpiece geometry. A finger contour that scores well across all workpieces simultaneously must generalize across diverse geometric profiles, making the problem increasingly challenging as more workpieces are added[5].

Together, these two problems provide a structured experimental basis for comparing optimization algorithms across varying objective space dimensionalities and objective landscape characteristics, within the practically relevant domain of robotic gripper design.

1.2 Goal and Tasks of the Thesis

The primary goal of this thesis is to **apply and comparatively evaluate optimization algorithms** spanning evolutionary multi-objective methods and Bayesian optimization for solving gripper configuration problems formulated as black-box multi-objective optimization tasks[8, 9]. The two gripper types, a multi-suction-cup gripper and a two-finger gripper, serve as well-defined benchmark problems on which the algorithms are systematically compared with respect to solution quality, convergence behaviour, and robustness.

To achieve this goal, the following tasks are defined:

- **Algorithm application and comparison:** Application and systematic comparison of a selected set of multi-objective optimization algorithms drawn from the available literature [10], evaluated under controlled experimental conditions using well-defined performance indicators across both gripper types and workpieces[11].
- **Workpiece selection and experimental coverage:** Selection of a diverse set of workpieces spanning a range of geometric properties including varying surface curvatures, sizes, and shapes to ensure that algorithm comparisons are not biased towards a specific object geometry. Workpiece diversity is treated as a deliberate experimental design choice to expose differences in algorithm robustness and generalization across objective landscapes of varying complexity.
- **Problem formulation:** Formulation of both gripper configuration problems as multi-objective optimization problems, including the definition of decision variables, objective functions, and constraints. For the suction-cup gripper, the objectives consist of the support triangle area and the grasp diversity evaluated per workpiece. For the two-finger gripper, each workpiece contributes one objective representing the conformance quality of the finger contour to that workpiece geometry.
- **Evaluation pipeline development:** Development of dedicated black-box evaluation pipelines for each gripper type, capable of assessing grasp quality given a gripper configuration and a target workpiece, and of serving as the objective function interface for all optimization algorithms.
- **Optimization framework implementation:** Implementation of a generalized optimization framework capable of handling multi-objective problems of varying dimensionality and interfacing with both evaluation pipelines through a unified black-box interface.
- **Result analysis:** Analysis and interpretation of algorithm performance across both gripper types and all workpieces, with particular focus on convergence behaviour, Pareto front

quality, and the sensitivity of algorithm rankings to workpiece geometry and objective space dimensionality.

2 Literature Review

2.1 Theoretical Background

2.1.1 Optimization Theory

Black-Box Optimization

A black-box optimization problem is one in which the objective function has no known analytical form and no gradient information is available to the optimizer[12]. The function can only be queried at candidate input points, returning a scalar or vector output. This setting arises naturally in engineering design when objectives are evaluated through computationally expensive procedures, making gradient computation either impossible or impractical. Because each function evaluation may be costly, black-box optimizers must find high-quality solutions within a limited evaluation budget, placing emphasis on sample efficiency[13]. The gripper configuration problems addressed in this thesis fall squarely within this category: objective values are computed through a series of geometric and numerical calculations applied to the candidate gripper configuration, and no closed-form expression exists that directly maps the decision variables to either the contact quality score or the grasp diversity metric.

Multi-Objective Optimization

Multi-objective optimization (MOO) refers to the simultaneous optimization of two or more conflicting objective functions over a shared decision space[14]. Formally, a multi-objective minimization problem is defined as:

$$\min_{\mathbf{x} \in \mathcal{X}} \mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_k(\mathbf{x})) \quad (2.1)$$

where $\mathbf{x}$ is the decision variable vector, $\mathcal{X}$ is the feasible decision space, and $k \geq 2$ is the number of objectives. When objectives conflict, no single solution minimizes all objectives simultaneously. Instead, a set of trade-off solutions exists, collectively forming the Pareto front. The goal of a multi-objective optimizer is therefore not to find a single best solution but to approximate this trade-off surface as accurately and completely as possible[14, 15].

Pareto Optimality

The concept of Pareto optimality provides the theoretical foundation for comparing solutions in a multi-objective setting. A solution $\mathbf{x}^{(a)}$ is said to *dominate* $\mathbf{x}^{(b)}$, written $\mathbf{x}^{(a)} \prec \mathbf{x}^{(b)}$, if and only if:

$$f_i(\mathbf{x}^{(a)}) \leq f_i(\mathbf{x}^{(b)}) \quad \forall i \in \{1, \dots, k\} \quad \text{and} \quad f_j(\mathbf{x}^{(a)}) < f_j(\mathbf{x}^{(b)}) \quad \text{for at least one } j \quad (2.2)$$

A solution is called *Pareto optimal* if no other feasible solution dominates it. The set of all Pareto-optimal solutions in the decision space is the *Pareto set*, and its image in the objective space is the *Pareto front*. In practice, algorithms compute an approximation of the Pareto front, and the quality of this approximation is assessed using performance indicators such as the hypervolume indicator[11, 16].

Evolutionary Algorithms

Evolutionary algorithms (EAs) are a family of population-based metaheuristics inspired by the principles of biological evolution. A population of candidate solutions is maintained across discrete generations. At each generation, new candidate solutions are generated through variation operators typically crossover, which recombines pairs of existing solutions, and mutation, which introduces random perturbations and a selection mechanism retains the most promising individuals for the next generation based on their objective values. This loop continues until a termination criterion is met, such as a maximum number of function evaluations or convergence of the population[17].

EAs are well suited to black-box multi-objective optimization because they require no gradient information, naturally maintain a diverse population of trade-off solutions, and are robust to noisy or discontinuous objective landscapes. Multi-objective evolutionary algorithms (MOEAs) extend the basic EA framework with selection mechanisms based on Pareto dominance or decomposition, enabling them to approximate the Pareto front in a single run[18, 19].

Bayesian Optimization

Bayesian optimization (BO) is a sequential model-based strategy for the global optimization of expensive black-box functions. A probabilistic surrogate model most commonly a Gaussian process (GP) is fitted to the observations collected so far[20]. The GP provides both a mean prediction and an uncertainty estimate at unobserved locations, which together define an *acquisition function* that balances exploration of uncertain regions with exploitation of promising ones[20, 21]. The point that maximizes the acquisition function is selected as the next evaluation, after which the surrogate is updated and the process repeats[22].

Hyperparameter Tuning

Optimization algorithms expose a set of hyperparameters such as population size, crossover probability, and mutation rate that control their search behaviour but are not adapted during the optimization run itself. The choice of hyperparameters can substantially affect algorithm performance, and using default values may systematically favour or disadvantage certain algorithms in a comparative study[23, 24].

Hyperparameter tuning addresses this by searching the hyperparameter space for settings that improve algorithm performance on the problem at hand before the main optimization run begins.

In this thesis, algorithm hyperparameters are initialized from the default values provided by the `pymoo` framework[10]. Where the optimization pipeline is configured to do so, a Bayesian optimization procedure is executed prior to the main run to replace these defaults with problem-specific settings, reducing the risk of performance bias from incidental parameter choices[25].

2.1.2 Multi-Objective Evolutionary Algorithms

The candidate algorithms evaluated in this thesis are all members of the multi-objective evolutionary algorithm (MOEA) family. Each follows the general EA loop described in Section 2.1.1 but differs in how it guides selection toward a well-distributed approximation of the Pareto front. Brief introductions to each algorithm are given below.

NSGA-II

The Non-dominated Sorting Genetic Algorithm II (NSGA-II) is among the most widely adopted MOEAs. Its selection mechanism is based on two criteria applied in sequence. First, the entire population is sorted into non-domination ranks: all non-dominated solutions receive rank 1, they are removed, and the process repeats until every individual has a rank. Second, within the same rank, solutions are ordered by *crowding distance*, a density estimate that measures how isolated a solution is from its neighbours in the objective space. Solutions with a higher crowding distance are preferred, encouraging spread along the Pareto front. NSGA-II uses an elitist strategy in which the combined parent and offspring population of size $2N$ is truncated to N using these two criteria. Its computational complexity per generation is $\mathcal{O}(kN^2)$, where k is the number of objectives and N is the population size[26].

NSGA-III

NSGA-III extends NSGA-II to problems with three or more objectives (many-objective problems), where crowding distance becomes an unreliable diversity measure. Instead of crowding distance, NSGA-III uses a set of structured *reference points* distributed uniformly on a normalized hyperplane in the objective space. Solutions are associated with their closest reference point, and selection preference is given to solutions that are associated with under-represented reference points and lie closest to their reference line. This mechanism explicitly promotes diversity across the objective space and has been shown to outperform NSGA-II on problems with three or more objectives[27].

MOEA/D

MOEA/D (Multi-Objective Evolutionary Algorithm based on Decomposition) takes a fundamentally different approach from Pareto-ranking methods. The multi-objective problem is decomposed into N scalar subproblems using a set of evenly distributed weight vectors and an aggregation function such as the weighted Chebyshev scalarization. Each subproblem is optimized simultaneously, with solutions sharing information within a defined neighbourhood of weight vectors. This neighbourhood-based collaborative search makes MOEA/D computationally efficient and well suited to problems where the Pareto front has a regular structure. Its complexity per generation is lower than NSGA-II, as it avoids global non-dominated sorting[10].

SMS-EMOA

The S-Metric Selection Evolutionary Multi-Objective Algorithm (SMS-EMOA) uses the hypervolume indicator directly as its selection criterion. It operates as a steady-state algorithm: at each step, a single offspring is generated, added to the population, and the individual that contributes the least to the population hypervolume is removed. This ensures that the population always converges toward solutions that maximize the dominated hypervolume with respect to a reference point. SMS-EMOA is particularly well suited to problems where solution quality is to be measured by the hypervolume indicator, as its selection pressure is directly aligned with that metric[28].

CMOPSO

The Competitive Multi-Objective Particle Swarm Optimizer (CMOPSO) adapts the particle swarm optimization (PSO) framework to the multi-objective setting. In standard PSO, each particle moves through the decision space guided by its personal best position and a global best position. CMOPSO introduces a competitive mechanism in which particles compete in pairs within the swarm; the losing particle updates its velocity by learning from the winning particle rather than from a fixed global best. An external archive of non-dominated solutions is maintained to provide guidance and preserve diversity across the Pareto front. This competitive selection strategy avoids premature convergence to a single region of the Pareto front and has been shown to achieve fast convergence while remaining competitive with dominance-based MOEAs on standard benchmark problems[29].

2.1.3 Gripper Theory and Geometric Representation

Grasp Theory

A grasp is a configuration of contact points between a gripper and a workpiece such that the gripper can resist external forces and torques applied to the object[30]. The classical criterion for grasp quality is *force closure*: a grasp is force-closed if the contact forces available at the contact points can produce any net wrench on the object, meaning that arbitrary disturbances can be resisted[31]. Force closure requires a sufficient number of contacts with appropriate friction properties and geometric placement.

Suction Cup Gripper and Grasp Diversity

The *diversity* of a grasp set can be quantified by combining two complementary measures of spatial spread.

Geodesic Distance as a Diversity Measure. The geodesic distance between two points on a surface is the length of the shortest path between them that lies entirely on the surface. As a diversity measure, geodesic distance captures surface-level spread that Euclidean distance in ambient space does not reflect for curved geometries. A larger mean geodesic distance between grasp contact points indicates broader sampling of the workpiece surface, which is desirable for robustness to workpiece pose variation[32].

Spatial Diversity via Mean Pairwise Euclidean Distance. The spatial spread of a grasp set can be quantified by the mean pairwise Euclidean distance between grasp positions in three-dimensional space, normalized by a characteristic object scale s to ensure comparability across objects of different sizes. The characteristic scale is defined as the Euclidean norm of the axis-aligned bounding-box extents:

$$s = \|\mathbf{e}\|_2 \quad (2.3)$$

where $\mathbf{e} \in \mathbb{R}^3$ contains the bounding-box side lengths. The normalized spatial diversity is then:

$$D_{\text{pos}}(\mathbf{p}) = \frac{1}{s} \cdot \frac{1}{n(n-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^n \|\mathbf{y}_i - \mathbf{y}_j\|_2 \quad (2.4)$$

where $\mathbf{y}_i \in \mathbb{R}^3$ is the position of the i -th grasp contact point and n is the total number of grasps considered. Normalizing by s ensures that the metric is scale-invariant, so that diversity values remain comparable across workpieces of differing sizes. This measure complements the geodesic measure by reflecting absolute spatial separation rather than surface-constrained distance.

Two-Finger Gripper and Contour Representation

The two-finger gripper applies gripping forces through two opposing jaw surfaces whose contour geometry determines how well the gripper conforms to the workpiece. Designing the jaw contour requires a representation that is compact enough to serve as a low-dimensional decision space for optimization, yet expressive enough to capture smooth, physically realizable shapes. Spline parameterization is a well-established approach for this purpose.

B-Splines. A B-spline curve of degree p is defined by a set of $n+1$ control points $\{\mathbf{P}_0, \mathbf{P}_1, \dots, \mathbf{P}_n\}$ and a non-decreasing knot vector $\mathbf{t} = \{t_0, t_1, \dots, t_{n+p+1}\}$ [33]. The curve is expressed as:

$$\mathbf{C}(u) = \sum_{i=0}^n N_{i,p}(u) \mathbf{P}_i, \quad u \in [t_p, t_{n+1}] \quad (2.5)$$

where $N_{i,p}(u)$ are the B-spline basis functions of degree p , defined recursively by the Cox-de Boor recurrence relation[33]. B-spline curves possess the *local support* property: moving a single control point affects the curve only in its neighbourhood, making them well suited to constrained shape optimization[34, 33]. They also guarantee C^{p-1} continuity everywhere except at repeated knots, ensuring smooth contour shapes without explicit smoothness constraints in the optimizer.

Spline-Based Shape Parameterization. Spline-based shape parameterization represents a geometric contour through a small set of control points, reducing an otherwise infinite-dimensional

shape space to a finite-dimensional parameter vector. Boundary control points are typically fixed to satisfy geometric constraints, while interior control points are free to vary and serve as the design variables. The number of free control points governs the trade-off between geometric expressiveness and search space dimensionality: more control points allow more complex shapes but increase the dimensionality of the optimization problem[34, 35].

2.2 State of the Art

Current research has made substantial progress in robotic grasp planning, adaptive gripper design, and task-specific end-effector optimization. In multi-suction grasping, recent methods focus on grasp pose selection, cup activation strategies, and motion safety, but typically assume a fixed suction layout and do not systematically study how cup arrangement affects the trade-off between stability and grasp availability [36] [37] [38]. In two-finger grasping, prior work has addressed compliant, soft, and underactuated finger designs, but direct optimization of rigid jaw geometry using workpiece-dependent, collision-aware metrics remains less explored [39] [40] [41]. Furthermore, none of these works conduct a comparative evaluation of optimization algorithms for the respective problem class.

This thesis addresses these gaps by treating gripper geometry as the primary optimization variable for both end-effector types, and by embedding the evaluation within a black-box multi-objective framework that enables systematic algorithm benchmarking across a diverse set of workpieces. The contribution lies not only in the problem formulations themselves, but in the rigorous comparison of optimization algorithms applied to them.

3 Methodology

This chapter describes the methodological decisions, theoretical frameworks, and evaluation strategies underlying the gripper configuration optimization study. It covers the problem formulation, the choice of optimization library and algorithms, the evaluation architecture, the workpiece selection rationale, and the comparative experimental design. Implementation details and experimental procedures are treated separately in the approach chapter.

3.1 Research Design

3.1.1 Research Paradigm

The thesis adopts a computational experimental methodology. Both gripper configuration problems are approached through mathematical modeling, computational-based evaluation, and controlled comparative optimization experiments. This design is appropriate because the objectives governing gripper quality cannot be expressed in closed analytical form; they depend on geometric interactions between the gripper and the workpiece that must be assessed computationally. The core methodological contribution is not the development of new optimization algorithms, but the rigorous comparative evaluation of existing algorithms on well-defined, practically motivated black-box problems.

3.1.2 Study Structure

The two gripper types are treated as independent optimization studies. The suction-cup gripper optimization seeks a cup layout that simultaneously maximizes grasp stability and grasp set diversity. The two-finger gripper optimization seeks a jaw contour that achieves high contact quality across a set of representative workpieces; since the evaluation pipeline computes a separate proximity-based score for each workpiece, each workpiece contributes one objective, making this a multi-objective problem whose dimensionality scales with the number of workpieces selected. In both cases, the problem structures and objective spaces differ, and the algorithm comparison is therefore conducted separately for each gripper type with results not aggregated across them.

3.1.3 Experimental Setup

For each problem, a fixed set of candidate algorithms is applied to a set of representative workpieces. Each algorithm is executed once per workpiece under identical conditions, including a shared evaluation budget and consistent initialization settings[42]. Algorithm performance is then analyzed and interpreted across all workpieces and both gripper types, with particular focus on three aspects: convergence behaviour over the evaluation budget, Pareto front quality as measured by established performance indicators, and the sensitivity of algorithm rankings to

workpiece geometry and objective space dimensionality[11]. This design allows conclusions to be drawn about which algorithms are best suited to each problem class and how performance varies with problem instance characteristics[43].

3.1.4 Scope and Limitations

Since no repeated independent runs are performed, the study does not support statistical inference about algorithm performance distributions. Conclusions are therefore descriptive and comparative in nature, grounded in the observed results across the workpiece set rather than in probabilistic significance testing. Furthermore, the evaluation relies entirely on computational calculation; physical validation of the optimized gripper configurations is outside the scope of this work.

3.2 Problem Formulation

3.2.1 Multi-Objective Optimization Framework

Both gripper configuration problems are formulated as multi-objective optimization problems. Let $\mathbf{q} \in \mathcal{Q}$ denote the decision vector and let d denote the number of decision variables. The general problem is formulated as

$$\min_{\mathbf{q} \in \mathcal{Q}} \mathbf{f}(\mathbf{q}) = (f_1(\mathbf{q}), f_2(\mathbf{q}), \dots, f_m(\mathbf{q})), \quad (3.1)$$

where m is the number of objectives, subject to inequality constraints

$$g_j(\mathbf{q}) \leq 0, \quad j = 1, \dots, n_c, \quad (3.2)$$

and variable bounds

$$q_i^L \leq q_i \leq q_i^U, \quad i = 1, \dots, d. \quad (3.3)$$

Objectives that are naturally maximized are negated to maintain a consistent minimization framework throughout. A multi-objective formulation is preferred over a weighted scalar aggregate because it produces a Pareto-optimal solution set that makes trade-offs explicit and preserves design alternatives without requiring the prior specification of preference weights.

The number of objectives m is directly determined by the number of workpieces used in the evaluation. This design choice has two methodological motivations. First, it forces the optimizer to find gripper configurations that perform well across geometrically diverse workpieces simultaneously, rather than overfitting to a single object. Second, it allows the study to examine algorithm behaviour as the dimensionality of the objective space increases, since adding workpieces directly increases m .

3.2.2 Parameterization of Gripper Geometry

A fundamental methodological decision is how to represent gripper geometry as a finite-dimensional decision vector suitable for black-box optimization [44] [1]. For both gripper types, the represen-

tation must satisfy three competing demands: it must be expressive enough to capture meaningful geometric variation, compact enough to keep the search space tractable, and physically interpretable so that every point in the search space corresponds to a realizable gripper configuration [45].

For the suction-cup gripper, the most general representation would specify the full Cartesian coordinates of each cup independently, yielding a six-dimensional search space for three cups in the plane. However, this representation introduces redundancy through rigid-body symmetries and admits degenerate configurations such as collinear or overlapping cups. An alternative is to exploit the geometric structure of the problem: the primary geometric quantity governing stability is the area of the triangle formed by the three cups, which depends on the radial spread and the relative angular spacing of the cups rather than their absolute positions. This motivates a reduced polar parameterization that encodes the common radial distance and the three angular positions, resulting in a four-dimensional search space that eliminates translational and overall rotational redundancy while retaining the degrees of freedom that directly affect the stability and collision behavior of the layout.

For the two-finger gripper, the finger jaw contour must be represented in a way that allows smooth, physically realizable shapes to be generated from a small number of parameters [46]. A spline parameterization is adopted because it provides a compact, smooth, and easily constrained contour representation, with fixed boundary control points enforcing the geometric constraints at the finger tip and base while the remaining interior control points form the free decision variables. The number of free control points can therefore be chosen directly to balance geometric expressiveness against search space dimensionality.

3.2.3 Selection of Objective Functions and Evaluation Metrics

The choice of objective functions determines what constitutes a good gripper configuration and therefore directly shapes the optimization problem [1, 46]. For each gripper type, the objectives must reflect physically meaningful performance criteria while remaining computable from geometry alone, since no physical simulation or hardware trial is available during optimization.

For the suction-cup gripper, two competing criteria govern layout quality. Grasp stability under external disturbances is classically characterized by force-closure metrics or grasp wrench space volume, both of which require contact force modeling and friction assumptions[47]. Since the optimization operates on geometry alone and no contact force model is available during the search, the area of the support triangle formed by the three cup positions is used as the stability objective. A larger triangle area directly increases the moment arm resisting tipping disturbances, making it a geometrically motivated and parameter-free indicator of layout quality[48]. The second criterion is the availability of valid grasps on the target workpiece[49]. A simple count of collision-free grasp poses does not capture the practical utility of the grasp set, since many similar poses offer little additional coverage. Diversity-based metrics, which measure the spread of valid grasps in both position and orientation, are better indicators of operational flexibility and have been used in related work on grasp set evaluation[50]. A weighted combination of spatial and orientational diversity is therefore adopted, with both components normalized to enable consistent comparison across workpieces of different scales.

For the two-finger gripper, the design objective is to find a jaw contour that achieves close

contact with the workpiece across as many placements as possible without collision[44]. For the two-finger gripper, the design objective is to find a jaw contour that achieves close contact with the workpiece across as many placements as possible without collision. The evaluation is performed using a voxel-based convolution scoring function developed at Fraunhofer IPA, which evaluates finger-to-workpiece proximity over all candidate placements without requiring explicit grasp candidate enumeration, and excludes colliding placements by construction. This function produces a single scalar score per workpiece that directly serves as the optimization objective.

3.3 Optimization Library Selection

All optimization algorithms are implemented and executed using the `pymoo` framework. This choice is motivated by several methodological considerations. `pymoo` is one of the most widely used Python frameworks for multi-objective evolutionary optimization and provides a unified implementation of a large number of algorithms, which reduces implementation bias and ensures that observed performance differences reflect genuine algorithmic properties rather than differences in code quality or implementation choices. It includes standardized performance indicators such as hypervolume, IGD, and GD, as well as built-in constraint handling and visualization tools, which improve the reproducibility and comparability of results. Furthermore, `pymoo` is actively maintained, open-source, and thoroughly documented, making it easier for other researchers to replicate and extend the experiments conducted in this thesis [10].

3.4 Algorithm Selection

The selection of optimization algorithms from the set available in `pymoo` was guided by two criteria: suitability for the problem class and coverage of fundamentally different optimization paradigms.

Regarding suitability, the gripper configuration problems are continuous, constrained, black-box multi-objective optimization problems. This rules out a substantial portion of the algorithms available in `pymoo`. Many algorithms in the library are designed exclusively for single-objective optimization, including genetic algorithms, differential evolution, particle swarm optimization, CMA-ES, evolution strategies, and direct search methods, and are therefore not applicable. Several multi-objective algorithms are designed for specialized problem settings outside the scope of this thesis. These include algorithms for dynamic optimization, where objectives or constraints change over time, and preference-based algorithms, which require user-defined preference information to guide the search toward specific regions of the Pareto front. Since both gripper problems are static and the goal is to approximate the full Pareto front without incorporating decision-maker preferences, these specialized methods were excluded[10].

Among the remaining general-purpose multi-objective algorithms, the selected methods were chosen because they represent fundamentally different optimization strategies rather than incremental variations of the same concept:

- **NSGA-II** represents Pareto-dominance-based selection and serves as the standard baseline for multi-objective evolutionary optimization[26].

- **NSGA-III** extends the dominance-based approach through reference directions, making it better suited to problems with a larger number of objectives, as arises when more workpieces are included[27].
- **MOEA/D** represents decomposition-based optimization, solving a set of scalar subproblems rather than relying solely on Pareto dominance[51].
- **SMS-EMOA** represents indicator-based optimization, directly maximizing the hypervolume indicator and thereby emphasizing Pareto front quality[28].
- **CMOPSO** provides a swarm-intelligence-based approach, employing a competitive particle swarm mechanism to balance exploration and convergence[29].

Together, these five algorithms cover the major paradigms of general-purpose multi-objective optimization, enabling a comparison of fundamentally different search strategies on the same gripper configuration problems[52, 53, 54].

In addition to the evolutionary algorithms, multi-objective Bayesian optimization is included as a comparative strategy. Bayesian optimization is motivated by its sample efficiency in expensive black-box settings: a probabilistic surrogate model is maintained over the objective functions and an acquisition function is used to guide evaluations toward promising regions of the search space. Its inclusion allows the study to assess whether surrogate-based methods offer advantages over population-based methods under the evaluation budgets considered.

3.5 Workpiece Selection

The workpieces were selected to provide geometric diversity rather than to optimize performance for a particular object. Since the objective of this thesis is to compare optimization algorithms, it is essential that the benchmark problems exhibit different geometric characteristics and consequently different optimization landscapes[55, 56].

The **IPAIntermediatePlate** represents a simple industrial workpiece with large planar surfaces and regular geometry, reflecting many components commonly encountered in manufacturing and material handling. Such geometry typically provides a relatively large feasible grasping region and a smoother objective landscape. In contrast, the **Stanford Bunny** exhibits a highly irregular three-dimensional shape with varying surface curvature and complex local features, resulting in a more challenging optimization problem characterized by narrower feasible regions, stronger collision constraints, and a potentially more multimodal objective landscape[57].

By evaluating all algorithms on both workpieces simultaneously, the study reduces the risk of drawing conclusions that are specific to a single geometry and enables a more robust assessment of algorithm performance across problems of varying complexity. The use of multiple workpieces as separate objectives also means that the comparison implicitly assesses how well each algorithm handles increasing objective space dimensionality[55].

3.6 Evaluation Architecture

The evaluation of candidate gripper configurations is decoupled from the optimization process through a client-server architecture. This separation constitutes a deliberate methodological design choice that enforces a strict black-box interface between the optimization strategy and the objective function computation. The optimization algorithms operate solely on input-output pairs $(\mathbf{q}, \mathbf{f}(\mathbf{q}))$ without access to internal evaluation logic[58]. This architecture also ensures that any optimization algorithm conforming to the interface can be applied to either evaluation pipeline without modification of either component, supporting reproducibility and extensibility.

3.7 Comparative Experimental Design

A controlled comparison protocol is adopted to ensure that observed performance differences are attributable to the optimization methods rather than to differences in experimental conditions. All algorithms are evaluated on the same workpieces, gripper parameterizations, objective functions, feasibility conditions, and evaluation budget. Each algorithm is executed under multiple independent random seeds to assess not only mean performance but also variability and robustness across runs[59]. Algorithm hyperparameters are initialized from the default values provided by the `pymoo` framework [10]. Where the optimization pipeline is configured to do so, a structured hyperparameter tuning procedure is executed prior to the main optimization run, replacing the defaults with problem-specific settings and reducing the risk of performance bias from incidental parameter choices. [60].

3.8 Performance Indicators

Algorithm performance is assessed using standard multi-objective performance indicators. The primary indicator is the *hypervolume indicator*, which measures the volume of objective space dominated by the solution set with respect to a reference point, thereby jointly capturing convergence quality and Pareto front diversity[11]. Convergence profiles over function evaluations are also recorded to assess the efficiency of each algorithm within the given evaluation budget. Robustness is characterized by the distribution of indicator values across repeated independent runs[61].

4 Implementation

4.1 Implementation of Multi-Suction Cup Gripper Optimization

This chapter presents the optimization framework developed for a multi-suction cup gripper consisting of three suction cups arranged in a triangular configuration. The framework uses the multi-objective optimization library *pymoo* [10], while the evaluation of candidate gripper layouts is delegated to an external service through an API. This separation enables a modular architecture in which the optimization logic and the domain-specific grasp evaluation remain decoupled.

4.1.1 Reduced Parameterization of Multi-Suction Cup Gripper

The gripper consists of three suction cups whose positions must ultimately be represented in Cartesian space in order to perform grasp generation, collision checking, and objective computation. However, the Cartesian suction cup coordinates are not used directly as optimization variables. Instead, the layout is parameterized in a reduced form to constrain the search space and encode geometric structure.

All three suction cups are constrained to lie on a common circle in the gripper plane. Their configuration is therefore defined by one shared radius r and three angular parameters $\alpha_1, \alpha_2, \alpha_3$:

$$\mathbf{q} = [r, \alpha_1, \alpha_2, \alpha_3] \quad (4.1)$$

The angular parameters determine the positions of the suction cups on the circle, while the common radius determines their distance from the gripper center. For a fixed gripper height z_0 , the Cartesian coordinates of the suction cups are obtained through the polar-to-Cartesian transformation

$$\mathbf{c}_i = \begin{bmatrix} r \cos(\alpha_i) \\ r \sin(\alpha_i) \\ z_0 \end{bmatrix}, \quad i = 1, 2, 3. \quad (4.2)$$

The resulting Cartesian gripper configuration is written as

$$\mathbf{p} = [\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3]. \quad (4.3)$$

Thus, the optimization is carried out in the reduced 4-dimensional parameter space $\mathbf{q}$, while the grasp evaluation operates on the derived Cartesian coordinates $\mathbf{p}$.

This formulation reduces the dimensionality of the search space from six planar degrees of freedom,

$$[(x_1, y_1), (x_2, y_2), (x_3, y_3)] \in \mathbb{R}^6, \quad (4.4)$$

to four variables,

$$[r, \alpha_1, \alpha_2, \alpha_3]. \quad (4.5)$$

The reduction improves search efficiency and excludes many geometrically redundant or undesirable layouts while preserving sufficient flexibility for diverse triangular suction cup arrangements.

Assumption A triangular configuration of three suction cups is assumed to yield a stable grasp [48]. This is motivated by the fact that three spatially distributed contact points form a statically stable support polygon, enabling effective load distribution and resistance to external disturbances.

4.1.2 Decision Variables and Bounds for Multi-Suction Cup Gripper

The optimization is performed over the reduced decision vector

$$\mathbf{q} = [r, \alpha_1, \alpha_2, \alpha_3]. \quad (4.6)$$

All variables are treated as continuous. The angular variables are bounded by

$$\alpha_i \in [0, 2\pi), \quad i = 1, 2, 3, \quad (4.7)$$

and the common radius is bounded by

$$r \in [r_{\min}, r_{\max}]. \quad (4.8)$$

The upper bound $r_{\max}$ is determined from the bounding box of the largest workpiece considered during optimization, ensuring that the gripper footprint remains compatible with the object dimensions. The lower bound $r_{\min}$ is determined from the diameter of an individual suction cup, ensuring that the configuration remains physically realizable and that the suction cups do not collapse into an excessively compact arrangement.

During initialization, the angular variables are sampled randomly in the interval $[0, 2\pi)$ and the radius is sampled within $[r_{\min}, r_{\max}]$. These values are then evolved by the selected optimization algorithm.

4.1.3 Optimization Framework of Multi-Suction Cup Gripper

The problem is formulated as a multi-objective minimization problem and solved using the *pymoo* library. The user configures the optimization through a request specifying:

- Population size N
- Maximum number of generations $G_{\max}$
- Choice of optimization algorithm
- Number of objectives n_{obj}
- Decision variable definitions and bounds
- Evaluation endpoint URL

Supported Algorithms

The framework supports the following multi-objective evolutionary algorithms:

Table 4.1: Supported multi-objective optimization algorithms.

Identifier	Algorithm
NSGA2	Non-dominated Sorting Genetic Algorithm II [26]
NSGA3	Non-dominated Sorting Genetic Algorithm III [27]
MOEAD	Multi-Objective Evolutionary Algorithm based on Decomposition [51]
CMOPSO	Competitive Multi-Objective Particle Swarm Optimization
SMSEMOA	S-Metric Selection Evolutionary Multi-Objective Algorithm [28]

For algorithms requiring reference directions, such as NSGA-III and MOEA/D, uniform reference directions are generated using the method of Das and Dennis [62]:

$$\Lambda = \text{get_reference_directions}(\text{"das-dennis"}, n_{\text{obj}}, n_{\text{partitions}}) \quad (4.9)$$

4.1.4 Initialization of Multi-Suction Cup Gripper Problem

An initial population of N candidate solutions is generated by sampling the reduced decision variables within their bounds:

$$\mathcal{P}^{(0)} = \{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_N\}. \quad (4.10)$$

Each candidate decision vector $\mathbf{q}_k$ is then transformed into a Cartesian gripper configuration $\mathbf{p}_k$ before objective evaluation.

4.1.5 Objective Function Evaluation of Multi-Suction Cup Gripper via API

The evaluation of objective values is decoupled from the optimization loop and delegated to an external evaluation service. For each candidate solution $\mathbf{q}$, the corresponding Cartesian suction cup coordinates are first computed by transforming the radial and angular parameters into 3D positions. The resulting Cartesian gripper configuration is then passed to the evaluation endpoint, which executes the grasp evaluation procedure and returns the objective values.

This process can be written as

$$\mathbf{p} = T(\mathbf{q}), \quad \mathbf{f}(\mathbf{q}) = \mathbf{f}(T(\mathbf{q})) = \text{API_call}(\mathbf{p}) \rightarrow [f_1(\mathbf{p}), f_2(\mathbf{p})]. \quad (4.11)$$

where $T(\cdot)$ denotes the transformation from the reduced parameterization to Cartesian suction cup coordinates.

In the implementation, this transformation is carried out programmatically in the evaluation pipeline before grasp generation and objective computation. For a candidate radius and list of angles, the suction cup coordinates are constructed as

$$\mathbf{c}_i = (r \cos(\alpha_i), r \sin(\alpha_i), z_0), \quad i = 1, 2, 3. \quad (4.12)$$

This architecture enables:

- Separation of concerns between optimization logic and domain-specific evaluation
- Scalability through independent deployment of the evaluation service
- Reusability of the optimization framework for different problem domains

Grasp Evaluation

The evaluation service receives a Cartesian gripper configuration $\mathbf{p} = [\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3]$ and computes the associated grasp quality measures through the following procedure.

1. Candidate Grasp Generation

A finite set of candidate suction grasps is generated from a uniform sampling of the work-piece surface. Let

$$\mathcal{U} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_M\} \quad (4.13)$$

denote the sampled surface points, where M is the number of initial samples. Each sampled point defines one candidate grasp by aligning the tool center point of the gripper with the corresponding surface location and local surface orientation.

In the initial implementation, $M = 100$ surface points were sampled. However, this occasionally resulted in too few valid grasps after collision filtering, which led to unstable behavior of the diversity objective. To reduce this effect, the sampling density was increased to $M = 1000$. The denser sampling provides a richer set of grasp candidates and substantially decreases the likelihood of degenerate cases with very few valid grasps.

2. Collision Filtering

The gripper is modeled as a rigid 3D body with fixed height, forming a triangular prism whose base is defined by the three Cartesian suction cup positions contained in $\mathbf{p}$.

Let

$$\mathcal{G}_{\text{cand}} = \{g_1, g_2, \dots, g_M\} \quad (4.14)$$

denote the set of candidate grasps generated from the sampled surface points. After collision checking, only collision-free grasps are retained:

$$\mathcal{G}_{\text{valid}} = \{g \in \mathcal{G}_{\text{cand}} \mid g \text{ is collision-free}\}. \quad (4.15)$$

This step removes physically infeasible grasps and yields the set of valid grasp candidates used in subsequent evaluation.

3. Clustering of Grasps

The set $\mathcal{G}_{\text{valid}}$ often contains many grasps that are nearly identical in position and orientation. When the number of valid grasps $m = |\mathcal{G}_{\text{valid}}|$ exceeds a threshold `nb_grasps_clustering`, a farthest-point clustering procedure is applied to select a diverse reduced subset. Otherwise, all valid grasps are used directly.

Each grasp g_i is represented by a pose descriptor

$$\mathbf{z}_i = [\text{vec}(\mathbf{R}_i)^\top, \mathbf{t}_i^\top]^\top \in \mathbb{R}^{12}, \quad (4.16)$$

where $\mathbf{R}_i \in SO(3)$ is the orientation and $\mathbf{t}_i \in \mathbb{R}^3$ is the position. Two distances are used: the translational distance

$$d_{\text{pos}}(i, j) = \|\mathbf{t}_i - \mathbf{t}_j\|_2 \quad (4.17)$$

and the orientation distance

$$d_{\text{ori}}(i, j) = \frac{1}{9} \|\text{vec}(\mathbf{R}_i) - \text{vec}(\mathbf{R}_j)\|_2^2. \quad (4.18)$$

Cluster centers are selected greedily by farthest-point sampling in position space. The first center is the grasp with the largest mean translational distance to all others, and each subsequent center maximizes its minimum distance to the already selected centers. This promotes broad spatial coverage.

Optionally, orientation diversity is added within each center's local neighborhood. For a selected center m_k , grasps within a spatial radius δ_{intra} form a neighborhood, from which up to q_{intra} additional grasps are chosen by farthest-point sampling on d_{ori} , capturing distinct orientations at similar positions.

The resulting clustered subset

$$\mathcal{G}_{\text{clustered}} \subseteq \mathcal{G}_{\text{valid}} \quad (4.19)$$

is then used for diversity evaluation.

4. Triangle Area Calculation

The area of the triangle formed by the three suction cups is computed from their Cartesian coordinates. If the planar coordinates of the three cups are (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) , then

$$A(\mathbf{p}) = \frac{1}{2} |(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)|. \quad (4.20)$$

This objective favors gripper layouts with a wider support polygon, which is associated with improved grasp stability.

5. Diversity Metric Calculation

A diversity score $D(\mathbf{p})$ is computed to quantify how broadly the resulting valid grasp set is distributed in position and orientation.

Let the set used for diversity evaluation contain n grasps. If clustering is performed, then

$$n = |\mathcal{G}_{\text{clustered}}|, \quad (4.21)$$

otherwise

$$n = |\mathcal{G}_{\text{valid}}|. \quad (4.22)$$

Let the corresponding grasp positions be

$$\{\mathbf{y}_1, \dots, \mathbf{y}_n\} \quad (4.23)$$

and the corresponding orientation matrices be

$$\{\mathbf{R}_1, \dots, \mathbf{R}_n\}. \quad (4.24)$$

To normalize the spatial term with respect to object size, a characteristic object scale is defined as the Euclidean norm of the axis-aligned bounding-box extents:

$$s = \|\mathbf{e}\|_2, \quad (4.25)$$

where $\mathbf{e} \in \mathbb{R}^3$ contains the bounding-box side lengths.

The spatial component of diversity is computed as the mean pairwise Euclidean distance between grasp positions:

$$D_{\text{pos}}(\mathbf{p}) = \frac{1}{s} \cdot \frac{1}{n(n-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^n \|\mathbf{y}_i - \mathbf{y}_j\|_2. \quad (4.26)$$

The orientation component is computed from the mean pairwise geodesic distance between rotation matrices. For two grasp orientations $\mathbf{R}_a, \mathbf{R}_b \in SO(3)$, the relative rotation is

$$\Delta \mathbf{R} = \mathbf{R}_a^\top \mathbf{R}_b \quad (4.27)$$

and the associated geodesic distance is

$$d_{\text{geo}}(\mathbf{R}_a, \mathbf{R}_b) = \cos^{-1} \left(\frac{\text{tr}(\Delta \mathbf{R}) - 1}{2} \right). \quad (4.28)$$

For numerical stability, the argument of $\cos^{-1}$ is clipped to the interval $[-1, 1]$.

The normalized orientation diversity is then defined as

$$D_{\text{rot}}(\mathbf{p}) = \frac{1}{\pi} \cdot \frac{2}{n(n-1)} \sum_{1 \leq i < j \leq n} d_{\text{geo}}(\mathbf{R}_i, \mathbf{R}_j). \quad (4.29)$$

The final diversity score is computed as the equally weighted combination

$$D(\mathbf{p}) = \text{clip} (0.5 D_{\text{pos}}(\mathbf{p}) + 0.5 D_{\text{rot}}(\mathbf{p}), 0, 1). \quad (4.30)$$

Continuity considerations Clustering is only applied when the number of valid grasps exceeds the predefined threshold `nb_grasps_clustering`. If fewer valid grasps are available, clustering is skipped and the diversity metric is evaluated directly on the valid grasp set. Therefore, the omission of clustering does not by itself introduce a discontinuity in the objective function.

A discontinuity may arise only in the degenerate case where fewer than two valid grasps remain after collision filtering. Since both the spatial and orientation components of the diversity metric are based on pairwise distances, these terms cannot be evaluated meaningfully for $n < 2$. In that case, the corresponding terms are set to zero. As a result, small changes in the gripper configuration may cause an abrupt transition between a nonzero diversity score and a degenerate zero-valued diversity contribution.

To reduce the likelihood of this case, the number of initially sampled surface points was increased from 100 to 1000. The denser surface sampling produces a larger set of candidate grasps and makes it much less likely that fewer than two valid grasps remain after filtering. Consequently, the diversity metric is evaluated on a sufficiently rich grasp set in the vast majority of cases, yielding a more stable optimization objective in practice.

Objective Evaluation

After grasp evaluation, each candidate layout is assigned the objective vector

$$\mathbf{f}(\mathbf{q}) = [-A(\mathbf{p}), -D(\mathbf{p})], \quad \mathbf{p} = T(\mathbf{q}). \quad (4.31)$$

Here, $A(\mathbf{p})$ is maximized because a larger triangle area is associated with improved support and stability, and $D(\mathbf{p})$ is maximized because greater diversity of valid grasps is associated with improved robustness. Since *pymoo* assumes minimization by convention, both objectives are negated.

4.1.6 Optimization Loop for Multi-Suction Cup Gripper

The evolutionary optimization proceeds as follows:

$$\mathcal{P}^{(t+1)} = \mathcal{A}(\mathcal{P}^{(t)}, \mathbf{f}), \quad t = 0, 1, \dots \quad (4.32)$$

Here, $\mathcal{A}$ denotes the chosen algorithm performing selection, crossover, mutation, and survival operations on the population of candidate decision vectors.

4.1.7 Termination Criteria for Multi-Suction Cup Gripper Optimization

The optimization terminates when any of the following conditions is satisfied:

Table 4.2: Termination criteria.

Criterion	Symbol	Description
Design space tolerance	x_{tol}	Change in decision variables below threshold
Objective space tolerance	f_{tol}	Relative improvement in objectives below threshold
Maximum generations	G_{max}	Maximum number of generations reached
Maximum evaluations	E_{max}	Maximum number of function evaluations reached

The tolerances are evaluated over a sliding window of w generations.

4.1.8 Performance Monitoring for Multi-Suction Cup Gripper

The hypervolume indicator [63] is tracked at each generation via a callback mechanism:

$$\text{HV}^{(t)} = \text{HV}(\mathcal{F}^{(t)}, \mathbf{r}) \quad (4.33)$$

Here, $\mathbf{r} \in \mathbb{R}^{n_{\text{obj}}}$ is the reference point and $\mathcal{F}^{(t)}$ denotes the set of objective vectors at generation t . A monotonically increasing hypervolume indicates convergence toward the Pareto front.

4.1.9 Optional Hyperparameter Optimization for Multi-Suction Cup Gripper Problem

The framework optionally supports automated tuning of algorithm hyperparameters, such as crossover probability and mutation rate, using a meta-optimization approach. A `HyperparameterProblem` is constructed around the selected algorithm and evaluated over multiple random seeds. The mean hypervolume across runs serves as the meta-objective:

$$\min_{\theta} \left\{ -\frac{1}{|\mathcal{Z}|} \sum_{s \in \mathcal{Z}} \text{HV}(\mathcal{F}_s, \mathbf{r}) \right\} \quad (4.34)$$

The best hyperparameters are then applied to the main optimization run.

4.1.10 Output of Multi-Suction Cup Gripper Optimization

The optimization produces a set of Pareto-optimal solutions:

$$\mathcal{P}^* = \{\mathbf{q} \mid \nexists \mathbf{q}' \in \mathcal{P} : \mathbf{f}(\mathbf{q}') \preceq \mathbf{f}(\mathbf{q})\}. \quad (4.35)$$

The final output includes:

- The Pareto-optimal set $\mathcal{P}^*$ in the reduced decision space, with corresponding objective values
- The associated radial gripper parameterizations $\mathbf{q} = [r, \alpha_1, \alpha_2, \alpha_3]$
- The hypervolume progression $\{\text{HV}^{(0)}, \text{HV}^{(1)}, \dots, \text{HV}^{(T)}\}$
- Total optimization wall-clock time

These solutions represent optimal trade-offs between grasp diversity and triangle area, i.e., grasp robustness and stability.

4.2 Implementation of Two-Finger Gripper Optimization

This chapter presents the optimization framework developed for a multi-suction cup gripper consisting of three suction cups arranged in a triangular configuration. The framework uses the multi-objective optimization library *pymoo* [10], while the evaluation of candidate gripper layouts is delegated to an external service through an API. This separation enables a modular architecture in which the optimization logic and the domain-specific grasp evaluation remain decoupled.

4.2.1 Decision Variables and Bounds for Two-Finger Gripper Problem

4.2.2 Optimization Framework for Two-Finger Gripper Optimization

Supported Algorithms

4.2.3 Initialization of Two-Finger Gripper Problem

4.2.4 Objective Function Evaluation of Two-Finger Gripper via API

Grasp Evaluation

Objective Evaluation

4.2.5 Optimization Loop for Two-Finger Gripper

4.2.6 Termination Criteria for Two-Finger Gripper Optimization

The optimization terminates when any of the following conditions is satisfied:

The tolerances are evaluated over a sliding window of w generations.

Table 4.3: Termination criteria.

Criterion	Symbol	Description
Design space tolerance	x_{tol}	Change in decision variables below threshold
Objective space tolerance	f_{tol}	Relative improvement in objectives below threshold
Maximum generations	G_{max}	Maximum number of generations reached
Maximum evaluations	E_{max}	Maximum number of function evaluations reached

4.2.7 Performance Monitoring for Two-Finger Gripper

4.2.8 Optional Hyperparameter Optimization for Two-Finger Gripper Problem

The framework optionally supports automated tuning of algorithm hyperparameters, such as crossover probability and mutation rate, using a meta-optimization approach. A `HyperparameterProblem` is constructed around the selected algorithm and evaluated over multiple random seeds. The mean hypervolume across runs serves as the meta-objective:

$$\min_{\theta} \left\{ -\frac{1}{|\mathcal{Z}|} \sum_{s \in \mathcal{Z}} \text{HV}(\mathcal{F}_s, \mathbf{r}) \right\} \quad (4.36)$$

The best hyperparameters are then applied to the main optimization run.

4.2.9 Output of Two-Finger Gripper Optimization

The optimization produces a set of Pareto-optimal solutions:

$$\mathcal{P}^* = \{ \mathbf{q} \mid \nexists \mathbf{q}' \in \mathcal{P} : \mathbf{f}(\mathbf{q}') \preceq \mathbf{f}(\mathbf{q}) \}. \quad (4.37)$$

The final output includes:

- The Pareto-optimal set $\mathcal{P}^*$ in the reduced decision space, with corresponding objective values
- The associated radial gripper parameterizations $\mathbf{q} = [r, \alpha_1, \alpha_2, \alpha_3]$
- Optionally, the derived Cartesian gripper configurations $\mathbf{p} = T(\mathbf{q})$
- The hypervolume progression $\{\text{HV}^{(0)}, \text{HV}^{(1)}, \dots, \text{HV}^{(T)}\}$
- Total optimization wall-clock time

These solutions represent optimal trade-offs between grasp diversity and triangle area, i.e., grasp robustness and stability.

5 Validation

5.1 Validation of Results for Suction-Cup Gripper

5.2 Validation of Results for Two-Finger Gripper

6 Evaluation

6.1 Results for Suction-Cup Gripper

6.1.1 Results for Evolutionary Algorithms

6.1.2 Results for Bayesian Algorithms

6.2 Results for Two-Finger Gripper

6.2.1 Results for Evolutionary Algorithms

6.2.2 Results for Bayesian Algorithms

7 Summary

7.1 Conclusion

7.2 Limitations

7.3 Future Work

A Appendix

Alles was den Hauptteil unnötig vergrößert hätte, z. B. HW-/SW-Dokumentationen, Bedienungsanleitungen, Code-Listings, Diagramme

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